REVENUE ANALYTICS · UNIT ECONOMICS DIAGNOSTIC

Is the growth sustainable, or just expensive?

A diagnostic of growth quality across channel unit economics, cohort retention, segment and product profitability, customer concentration, and a bounded, stress-tested spend-reallocation scenario.

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Coverage 36 months, January 2023 to December 2025 · 9,000 customers · 69,950 transactions · 5.6M acquisition spend

Contents

1 Executive summary

Revenue grew 9.6-fold across the three-year window, comparing the first and last six months, but the growth is carried by the most expensive customers the business buys. The constraint is the spend mix, not demand. Moving the budget that already exists toward the channels that pay back adds $7.9M of annual contribution without raising the budget by a dollar.

$54.6M	$16.6M	20.3×	0.43×
TOTAL REVENUE	CONTRIBUTION MARGIN	BEST CHANNEL LTV/CAC	WORST CHANNEL LTV/CAC
36-month window	30.3% of revenue	organic · 1.8m payback	social ads · loses money

The headline is unambiguous. Monthly revenue rose from $65K in January 2023 to $3.1M in December 2025, and contribution margin tracked it closely in absolute dollars. A revenue chart on its own would read as an unqualified success. Five separate analyses say otherwise, and they agree.

First, channel economics split sharply with no middle ground. Organic, referral, and partners return between 4.4 and 20.3 times their acquisition cost and pay back in under nine months. Paid search and social ads return 0.93 and 0.43 times cost, so the average customer they buy is worth less than the cost to acquire them. These two channels hold $3.78M, or 68%, of total acquisition spend, while generating only 16% of channel contribution. The three efficient channels do the inverse: 25% of spend, 78% of contribution.

Second, growth is volume-led rather than monetization-led. Of the $15.1M revenue increase between the first and last six months, 69% comes from acquiring more customers and only 25% from higher revenue per customer. Volume bought through channels that lose money is the weakest form of growth, because holding the line depends on continued spend.

Third, retention decays fast and early. Median revenue retention falls to 68% by month 6 and 50% by month 12, with the steepest loss between months 3 and 6. Only 16.7% of cohorts show revenue expansion at month 6, so the median customer shrinks in their first half-year. The acquisition treadmill cannot be slowed until this early-life decay is fixed.

Fourth, margin concentrates in the least efficient places. Enterprise is the largest segment at 48% of revenue but carries the lowest margin rate at 26.2%, the only segment below the 30% quality floor. The Services product line runs at just 17.7% margin on $13.6M of revenue. Regional margins, by contrast, are banded within 1.0 points, so geography is not the problem. Mix and cost-to-serve are.

Fifth, revenue is highly concentrated. The top 10% of customers account for 64% of revenue and the top 20% for 81%, with a Gini coefficient of 0.77. That concentration makes retention of high-value customers a first-order risk, and it makes the early cohort decay more expensive than the median retention figure suggests.

The recommended response is a bounded reallocation. Scaling the three efficient channels within guardrails and pulling 35% from the two inefficient ones lifts estimated annual contribution from $16.6M to $24.5M, an uplift of $7.9M (48%), with no extra budget. The uplift stays positive in every stress case tested, from $2.2M in the worst case to $12.8M in the best, and across all 5 deterministic seeds it averages $8.0M with a coefficient of variation of 1.3%. The full set of recommendations and their priority order is in Section 7.

2 Context and objectives

Top-line revenue growth is the metric most often celebrated and least often interrogated. A revenue line that bends upward can hide three separate problems at once: customers acquired below the cost to serve them, cohorts that leak revenue faster than new cohorts replace it, and margin diluted by an unfavourable product or segment mix. None of these appear in a revenue chart. All three change what a business should do next.

This analysis exists to answer a budget question, not to populate a dashboard. The business is spending $5.6M a period to acquire customers and wants to know whether that spend is building a durable asset or renting a revenue line that stops the moment spend stops. The question is asked five ways, each with a method chosen to give a decision rather than a description.

QUESTION	METHOD	DECISION IT INFORMS
Is growth converting into margin?	Monthly revenue and contribution margin trend, margin rate vs floor	Whether to defend or improve margin
What is driving the growth?	Decomposition into volume, monetization, and mix	Where to invest for durable growth
Do cohorts hold their value?	Median activity and revenue retention by cohort age	Whether retention or acquisition is the lever
Which channels deserve budget?	LTV/CAC, payback, and spend-to-contribution alignment per channel	How to allocate acquisition spend
What is the upside from reallocating spend?	Bounded scenario with stress and seed-stability tests	Size and confidence of the reallocation

The scope is deliberately narrow. This is not a forecast, a market sizing, or an attribution study. It is a diagnostic of growth quality on the data available, built so that each finding either confirms the current allocation or changes it. Where a finding cannot be settled with the data on hand, Section 6 says so plainly rather than papering over the gap.

Who should read this

The primary audience is the executive or commercial owner deciding next quarter's acquisition budget. The secondary audience is the analytics or finance reviewer who needs to trust the numbers before acting on them. The report is written for the first audience in the prose and for the second in the methodology and appendix.

3 Data and methodology

The pipeline runs end to end from raw tables to validated outputs, with each stage writing deterministic files. Three raw tables feed it: 9,000 customers, 69,950 transactions, and 6,576 daily channel spend records, spanning January 2023 to December 2025. Customer-level metrics, the monthly revenue health series, the cohort table, channel unit economics, segment and product profitability, and the scenario outputs are all derived from these three tables by reproducible transformations.

Metric definitions

Contribution margin is revenue minus direct delivery cost. Observed lifetime value is the mean cumulative contribution margin per acquired customer, computed across the full base including the 843 customers (9.4%) who never transacted, so the figure is not inflated by survivorship. Customer acquisition cost is period-level channel spend divided by the customers acquired in that channel. The LTV-to-CAC ratio divides the two; payback is the number of months of contribution required to recover CAC.

A channel is classified efficient when LTV/CAC is at least 3.0 and payback is 12 months or less, inefficient when LTV/CAC falls below 1.0 or payback exceeds 24 months, and borderline in between. The margin quality floor is set at 30%. These thresholds live in a single metric registry that the analysis, the dashboard, the chart pack, and the validation gate all import, so no definition can drift between outputs.

Cohort construction

Customers are grouped into monthly signup cohorts. For each cohort, revenue retention at age m is that cohort's revenue in month m divided by its revenue in month 0, and activity retention is the share of the cohort still transacting. Reported curves use the median across cohorts at each age, so a single large or small cohort cannot dominate the read. Later-month retention figures draw only on cohorts old enough to have reached that age, which is stated wherever it matters.

Revenue decomposition

The growth decomposition compares the first six months of the window against the last six. The total revenue change is split into a customer-volume effect, an average-revenue (monetization) effect, and a segment-mix effect, with any unexplained difference carried as a residual. The residual is zero here, so the three effects fully account for the change. Segment is the mix dimension; other definitions of mix, such as region or product, would allocate the mix term differently, which Section 6 notes.

Scenario engine

The reallocation scenario is deliberately conservative. It applies bounded CAC and LTV elasticities to each channel, caps any channel's scale-up at 100% of current spend, and holds back budget rather than forcing it into channels that have run out of efficient headroom. The scenario is stress-tested across best, base, and worst elasticity cases, and repeated across 5 deterministic seeds to confirm the result is a property of the policy rather than of one random draw. The total budget is held constant in every case, so the uplift is a pure allocation effect.

Data quality gate

All 11 raw-data validation checks pass before any analysis runs: schema match, grain uniqueness, referential integrity, date coverage, channel-domain alignment, and value-range sanity. The gate flagged 258 transactions (0.37%) where cost exceeds revenue. That share sits below the 1% review threshold, so the rows are retained as genuine cost-to-serve exceptions rather than scrubbed. At the customer level, 22 customers carry a negative lifetime contribution margin, which the LTV figures absorb rather than exclude. The full gate is reproduced in Appendix F.

Limitations of the data

The data is synthetic. It is built to be realistic and internally consistent, which makes it suitable for demonstrating method and for relative comparison between channels, segments, and cohorts. It is not a forecast of any real market. Observed LTV reflects only the window available, so long-horizon cohort reads draw on mature cohorts only. These limits are revisited in Section 6.

4 Analytical framework

Growth quality is not a single metric. It is the answer to whether the next dollar of revenue is worth more, the same, or less than the average dollar already on the books. This report builds that answer from four layers, each one narrowing the question the previous layer leaves open.

The first layer is the **aggregate trajectory**: does revenue growth carry margin growth, and does the margin rate improve, hold, or erode as the business scales. A flat margin rate during rapid growth is the first signal that incremental revenue is no better than average, and it sets up every question that follows.

The second layer is **attribution of the growth**: whether the increase comes from more customers, more revenue per customer, or a shift in mix. Volume-led growth and monetization-led growth demand different responses. The decomposition separates them so the response can be matched to the cause.

The third layer is **unit economics and durability**: whether the customers being acquired are worth more than they cost, and whether they stay long enough to pay back. This is where channel LTV/CAC, payback, and cohort retention combine. A channel can look cheap on CAC and still destroy value if the customers it buys churn before payback.

The fourth layer is **concentration and structural risk**: where revenue and margin pool, and how exposed the business is to a small set of customers, segments, or products. Concentration is not inherently bad, but it changes which risks matter and which levers move the margin line.

The reallocation scenario sits on top of all four. It takes the channel economics from layer three, applies conservative elasticities, and tests how much contribution a budget-neutral move can add. The framework is built so that the scenario is a consequence of the findings, not an assertion bolted onto them.

> Read the findings as a single argument. Each section answers the question the previous one raises, and the reallocation in Section 5.8 is the action the first seven findings point to.

5 Findings

5.1 The growth trajectory is real, fast, and uneven

Monthly revenue rose from $65K in January 2023 to $3.1M in December 2025. Comparing like with like, the last six months averaged $2.8M a month against $293K in the first six, a 9.6-fold increase across 36 months. Contribution margin tracked revenue closely in absolute terms, which is the first thing to confirm: the business is not buying revenue at the expense of total margin dollars. Both lines climb together.

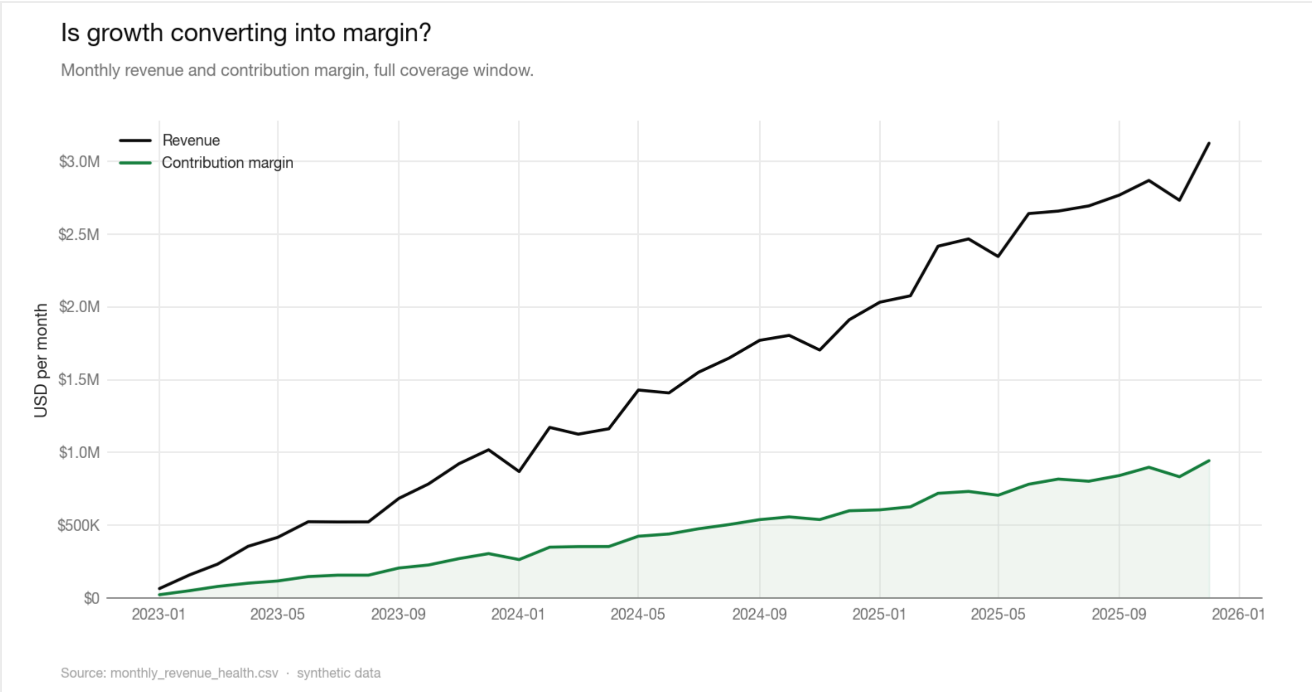

Figure 1. Revenue and contribution margin rise together in absolute dollars. The gap between the two lines, which is the margin rate, does not widen as the business scales.

The growth is not smooth. Month-on-month revenue change averages 13.9%, but the series includes 7 down months across the window, most of them in the seasonal January and mid-year dips visible in Figure 2. Durability shows up in how the business handles those down months, and here it recovers each time without breaking the trend. The volatility is seasonal noise around a strong underlying climb, not a structural wobble.

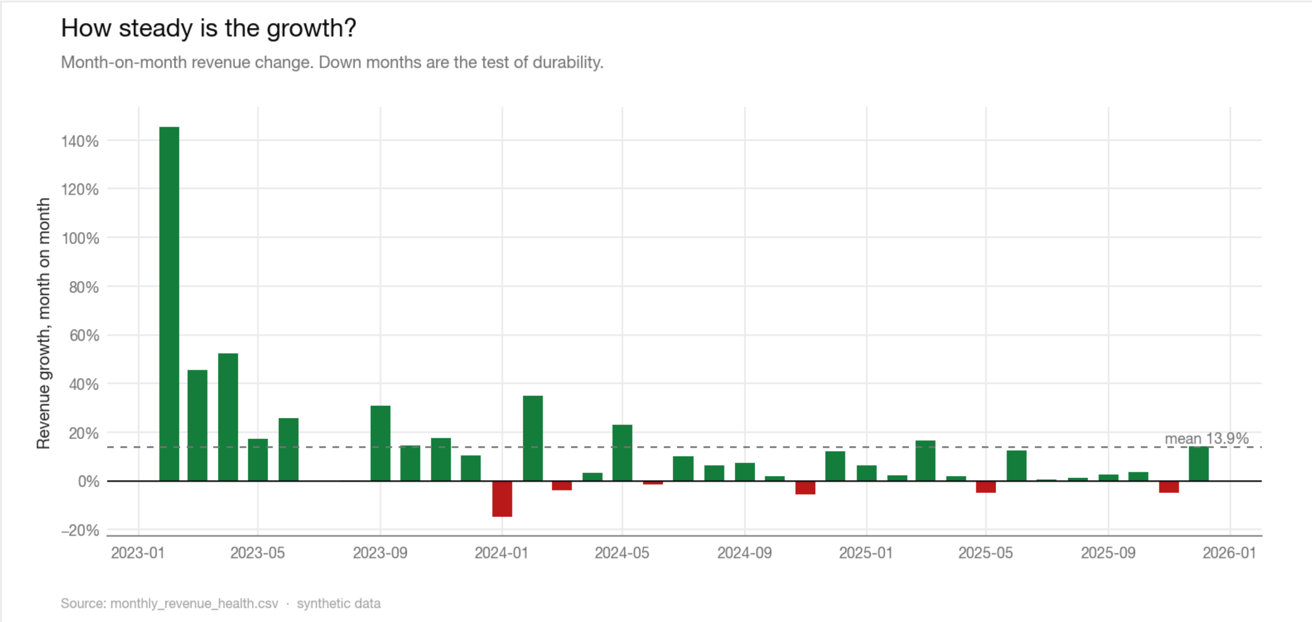

Figure 2. Month-on-month revenue growth. The 7 down months are shallow and seasonal, not a sign of a stalling trend.

The composition of that growth is the first warning. The active base grew from 67 customers in the opening month to 2,245 in the closing month, while revenue per active customer moved only from $976 to $1,392. The base is doing almost all the work; monetization per customer is close to flat. A business adding this many customers while holding revenue per customer steady is growing by addition, not by deepening the value of the relationships it already has.

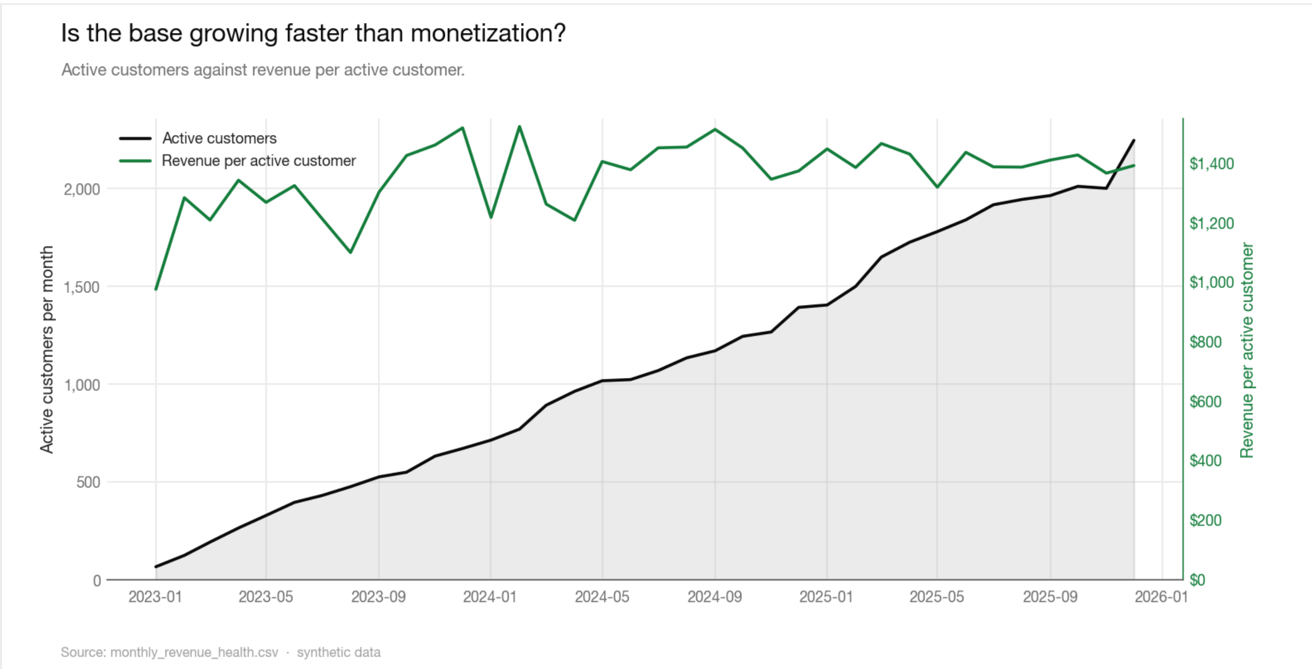

Figure 3. The active base climbs steeply while revenue per active customer stays in a narrow band. Growth is coming from more customers, not richer ones.

5.2 Margin quality is only holding, not improving

The qualifier on the trajectory is the rate. Contribution margin holds in a tight band between 28% and 35%, hovering at the 30% quality floor rather than improving as the business scales. A company adding this much revenue would normally expect operating leverage to lift the margin rate over three years. Here it does not move.

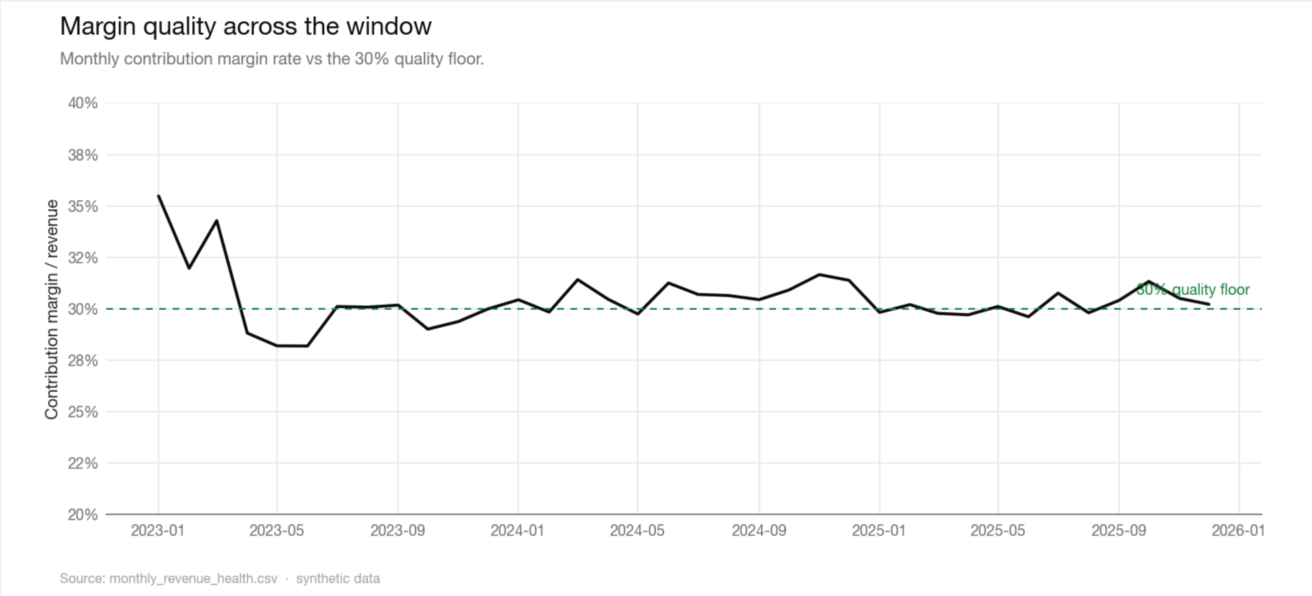

Figure 4. Monthly contribution margin rate against the 30% quality floor. The rate oscillates around the floor for the full window and shows no upward drift.

A flat margin rate during rapid growth carries a specific meaning: the incremental revenue is no better in quality than the average revenue already on the books. If the new customers were more profitable than the existing base, the rate would rise. If they were dragging, it would fall. It does neither, which says the business is cloning its average economics at scale rather than improving them. The next three findings explain why the rate is stuck, and the reallocation in Section 5.8 is the move that would unstick it.

5.3 Growth is volume-led, not monetization-led

Decomposing the $15.1M revenue change between the first and last six months separates three effects: more customers, more revenue per customer, and a shift in segment mix.

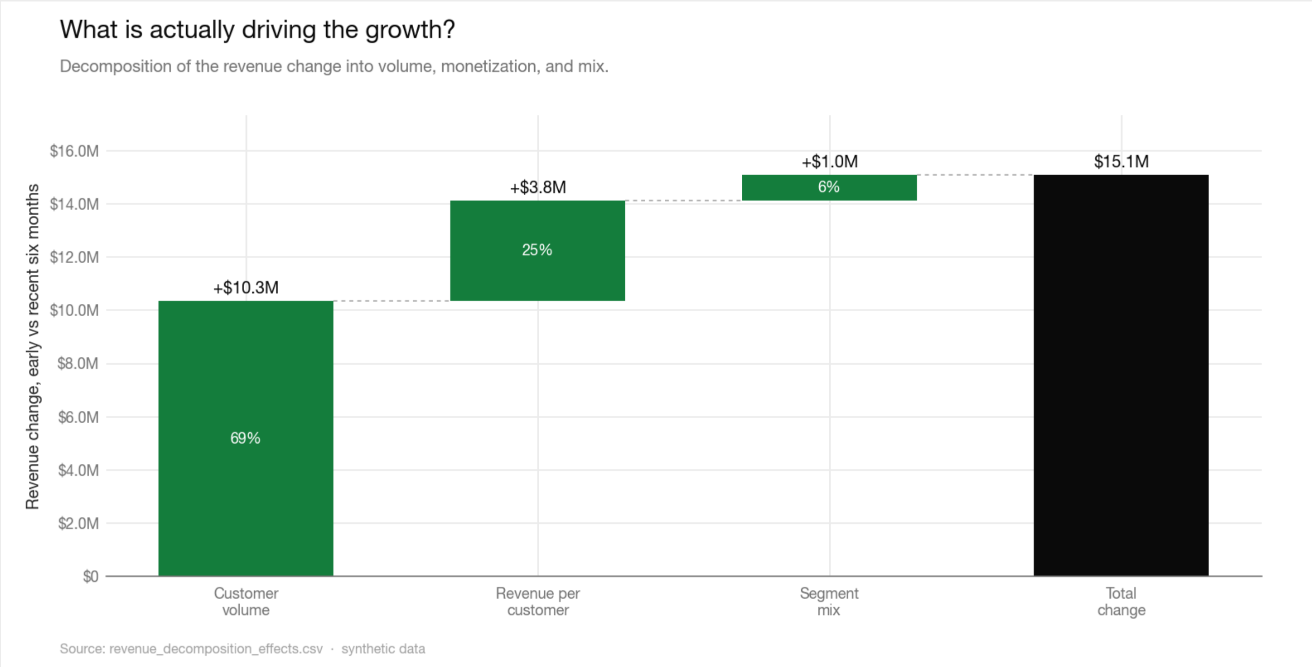

Figure 5. Volume accounts for 69% of the revenue change, monetization 25%, and segment mix 6%. The residual is zero, so the three effects fully account for the change.

EFFECT	CONTRIBUTION TO CHANGE	SHARE
Customer volume	$10,349,685	68.6%
Revenue per customer	$3,771,454	25.0%
Segment mix	$966,809	6.4%
Total change	**$15,087,948**	**100.0%**

Volume accounts for 69% of the increase. Monetization adds 25% and mix a further 6%. Volume-led growth is not wrong in itself, but it is only as good as the channels buying that volume. When most incremental customers arrive through channels that lose money, as Section 5.5 shows, volume growth becomes a recurring cost rather than a compounding asset. This is the mechanism behind the flat margin rate in Figure 4: the business is buying average-quality customers in bulk and the average is not improving.

The mix effect is small at 6%, which is its own finding. The business is not growing into a richer customer mix that would lift the rate on its own. If anything, the structural margin problem documented in Section 5.6 means a larger mix shift toward the biggest segment would pull the rate down, not up.

5.4 Cohorts decay fast in the first six months

If growth depends on volume, retention determines how much of that volume the business keeps. Median revenue retention falls to 88% by month 3 and 68% by month 6. Activity retention is worse, at 65% by month 6 and 42% by month 12.

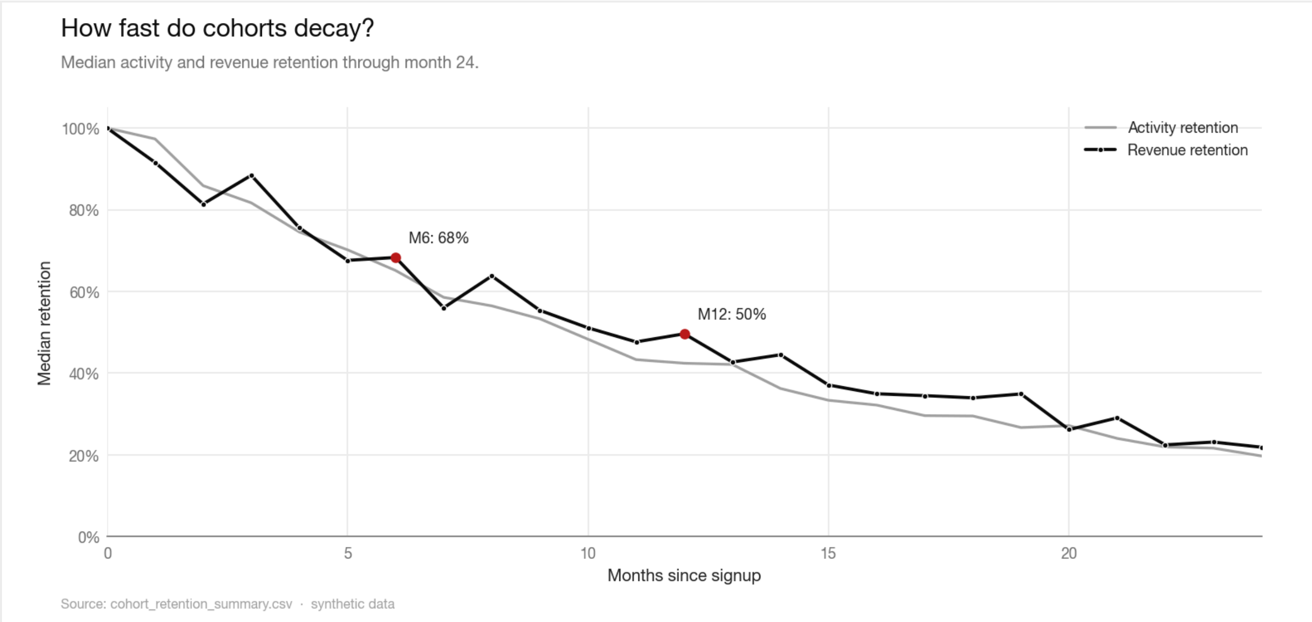

Figure 6. Median activity and revenue retention through month 24. The steepest loss is concentrated between months 3 and 6, the window where onboarding and first-value delivery matter most.

Revenue retention sits above activity retention at most ages, which says the customers who stay spend somewhat more, a partial offset to logo churn. It is not enough to change the conclusion. Only 16.7% of cohorts show revenue expansion at month 6, so the median customer is shrinking, not growing, in their first half-year. The cohort heatmap confirms that this is structural rather than a property of one or two unlucky cohorts.

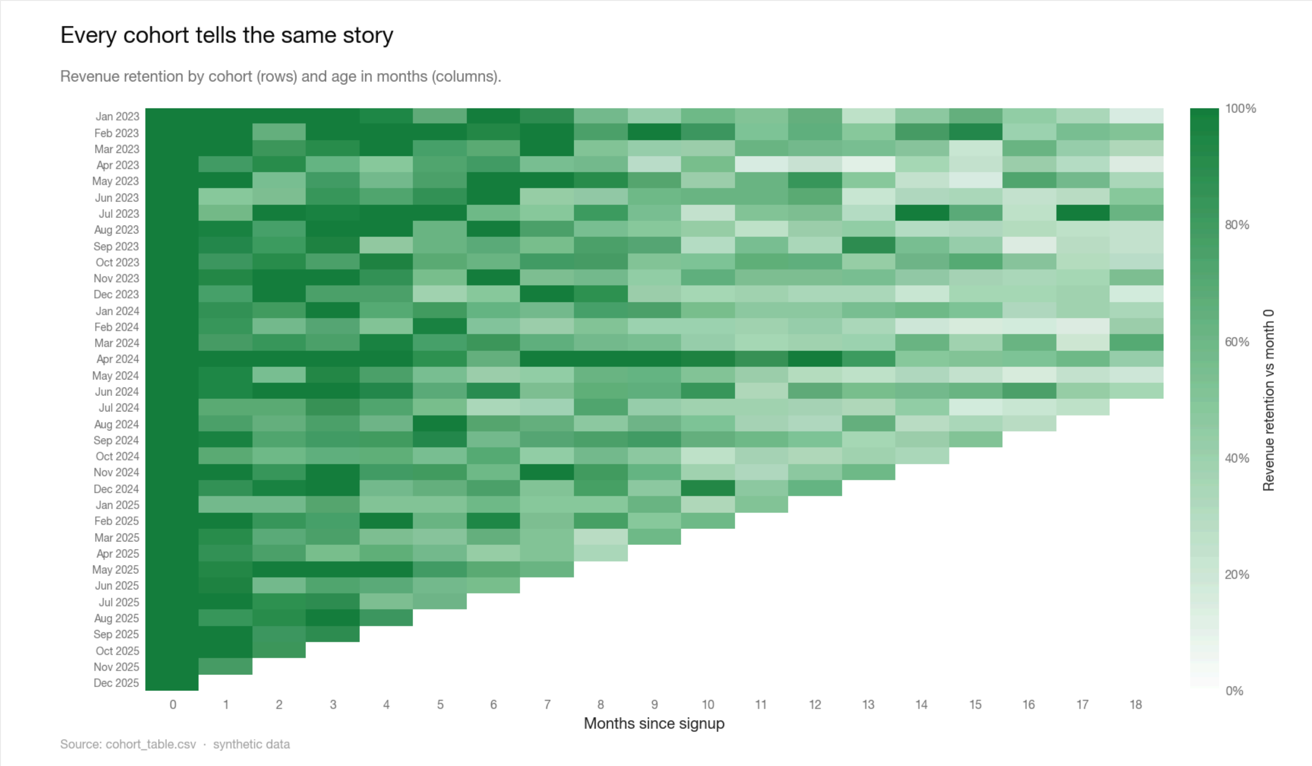

Figure 7. Revenue retention by cohort (rows) and age (columns). The colour fades along every row at a similar pace, so the decay pattern repeats across cohorts rather than being driven by a few outliers.

The combination is costly. The business acquires volume through expensive channels and then loses a large share of it before the customer pays back. Because revenue is concentrated in a small set of high-value customers, as Section 5.7 shows, losing the wrong customers early is more damaging than the median curve implies. Slowing the acquisition treadmill is not possible until this early-life decay is addressed, which is why retention is a named recommendation in Section 7 rather than a footnote.

5.5 Two channels destroy value and hold most of the budget

This is the central finding. Channel economics split into two groups with no overlap, and the budget is pointed at the wrong group.

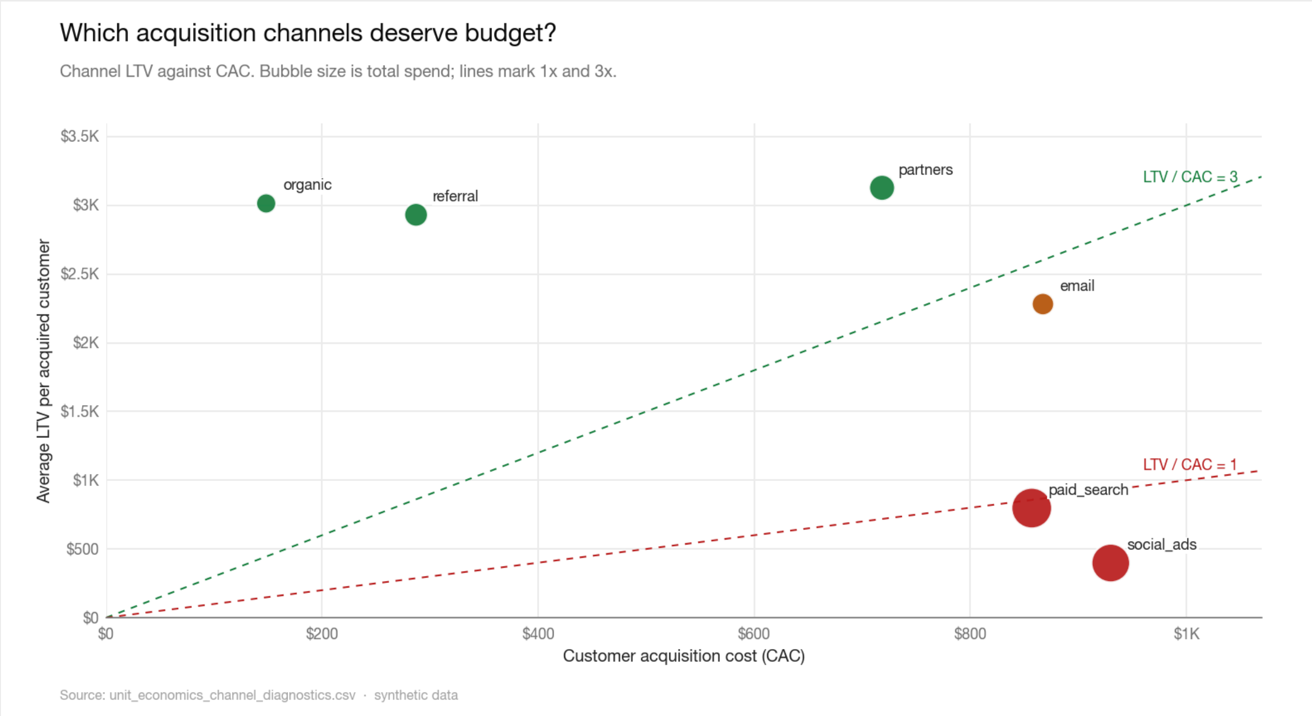

Figure 8. Channel LTV against CAC, with bubble size showing total spend. Three channels sit above the 3× line, two below 1×, and none in between. The two largest bubbles, paid search and social ads, sit in the value-destructive zone.

CHANNEL	CUSTOMERS	SPEND	CAC	AVG LTV	MED LTV	LTV/CAC	PAYBACK (MO)	STATUS
organic	1,736	$257,299	$148	$3,014	$984	20.33	1.8	**efficient**
referral	1,668	$478,649	$287	$2,931	$1,117	10.21	3.5	**efficient**
partners	889	$638,558	$718	$3,127	$1,085	4.35	8.3	**efficient**
email	455	$394,559	$867	$2,282	$742	2.63	13.7	**borderline**
paid_search	2,338	$2,003,250	$857	$797	$221	0.93	38.7	**inefficient**
social_ads	1,914	$1,779,919	$930	$398	$98	0.43	84.2	**inefficient**

Organic, referral, and partners clear the 3× bar with room to spare and pay back inside nine months. Paid search and social ads sit below the 1× line, meaning the average customer they buy is worth less than the cost to acquire them. The efficiency gap is not marginal. It is more than an order of magnitude between the best and worst channel.

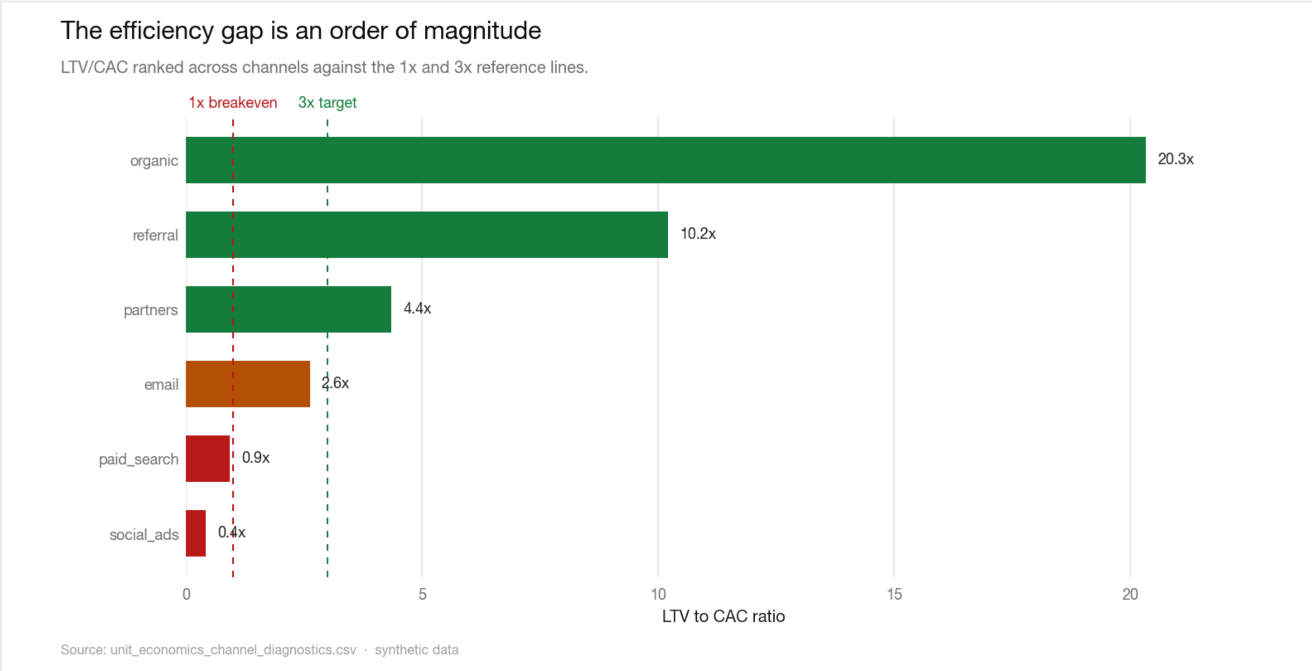

Figure 9. LTV/CAC ranked across channels. Organic returns 20× its cost; social ads returns 0.43×. Email is the only borderline case.

The problem is not that the inefficient channels exist; it is their weight. Together they hold $3.78M of acquisition spend, 68% of the total, while returning under a dollar of contribution per dollar spent. The misallocation is clearest when each channel's share of spend is set against its share of contribution.

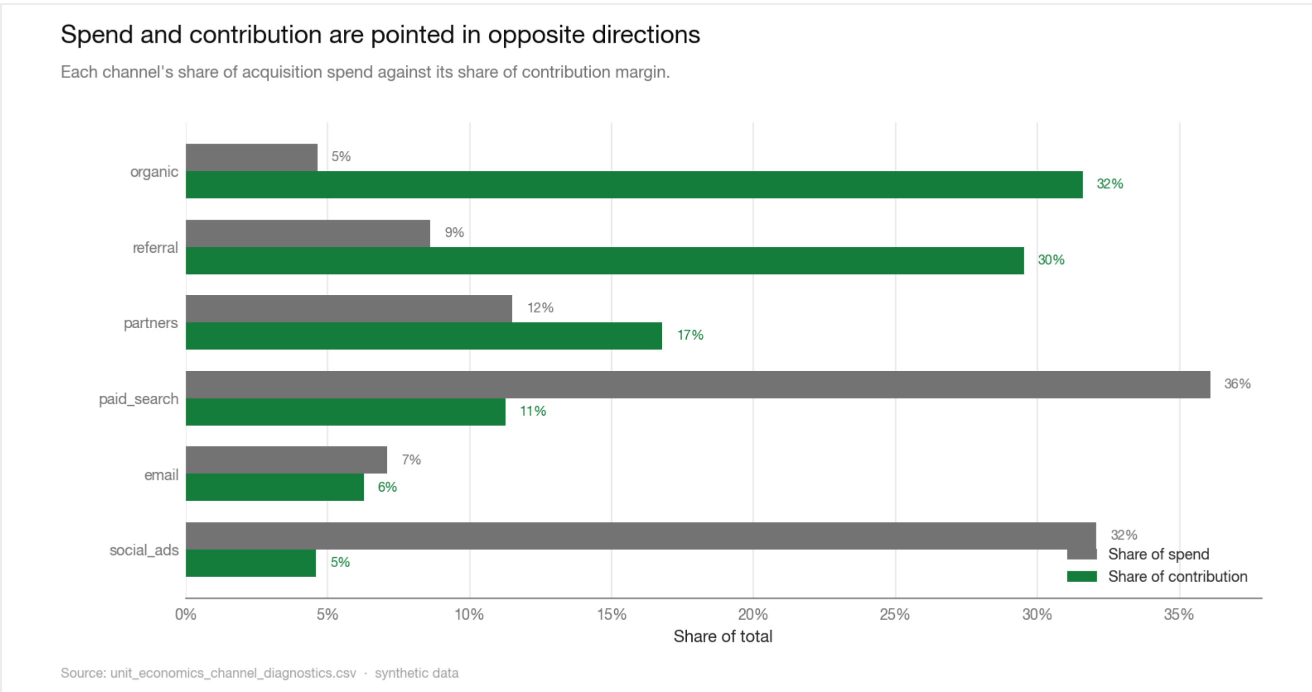

Figure 10. Share of spend against share of contribution by channel. Organic takes 5% of spend and produces 32% of contribution; paid search and social ads take 68% of spend and produce 16% of contribution.

Email is borderline at 2.63× with a 14-month payback. It is a hold-and-fix case rather than a scale-or-cut one. The median LTV column tells a sharper version of the same story than the mean: social ads has a median LTV of $98 against a CAC of $930, so the typical customer it buys is deeply underwater even before averaging in the rare high-value account.

> Reallocating budget away from these two channels is the single highest-value move available, and it requires no incremental spend. Section 5.8 sizes it.

5.6 Margin concentrates in the lowest-rate segment and product

Profitability by segment carries a structural risk that a revenue view hides. The largest segment is also the least efficient.

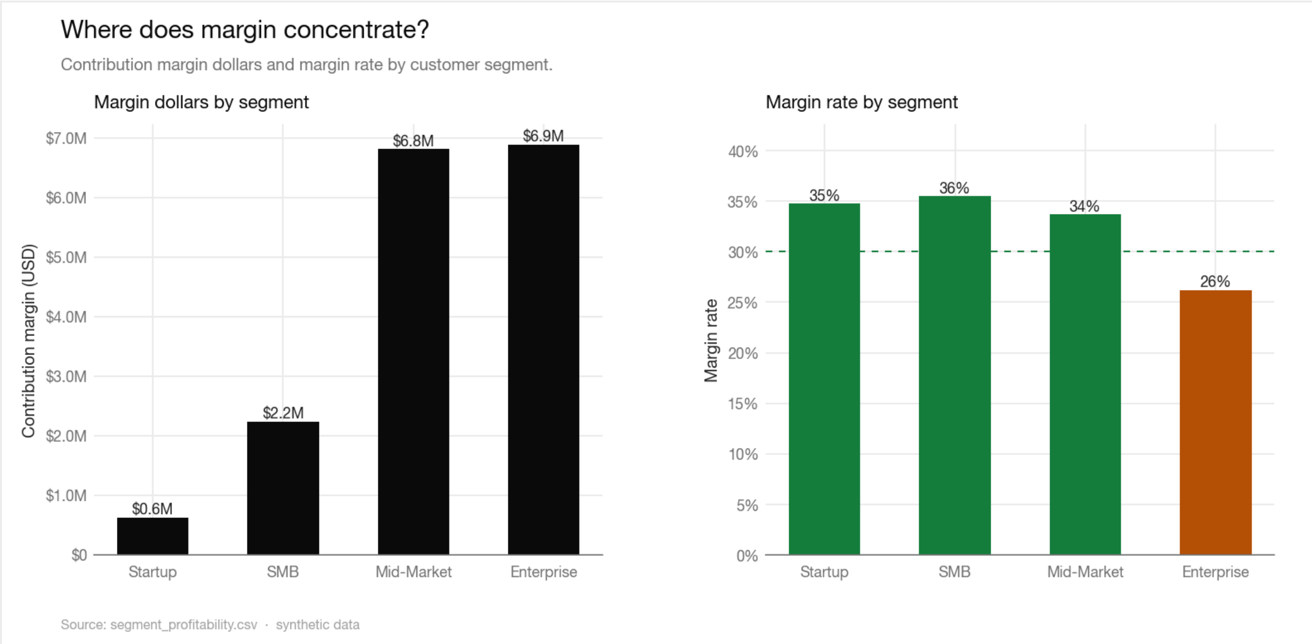

Figure 11. Contribution margin dollars and margin rate by segment. Enterprise carries the most margin dollars at the lowest rate, and it is the only segment below the 30% floor.

SEGMENT	CUSTOMERS	REVENUE	CONTRIBUTION MARGIN	MARGIN RATE	REVENUE SHARE
Enterprise	779	$26,282,659	$6,890,020	26.2%	48.1%
Mid-Market	2,415	$20,220,544	$6,814,179	33.7%	37.0%
SMB	3,184	$6,288,792	$2,232,644	35.5%	11.5%
Startup	2,622	$1,803,972	$627,187	34.8%	3.3%

Enterprise is the largest segment at 48% of revenue and $6.9M of margin dollars, yet it carries the lowest margin rate at 26.2%, below every smaller segment. Mid-Market, SMB, and Startup all run between 36% and 34%. Nearly half the revenue base sits in the least efficient segment, which is a concentration risk: pricing or cost-to-serve pressure in Enterprise hits the margin line harder than its rate alone suggests.

Geography, by contrast, is not where the margin problem lives. Regional margin rates are banded within 1.0 points of each other, from 29.9% in LATAM to 30.3% in North America. Regions differ in size, not in efficiency.

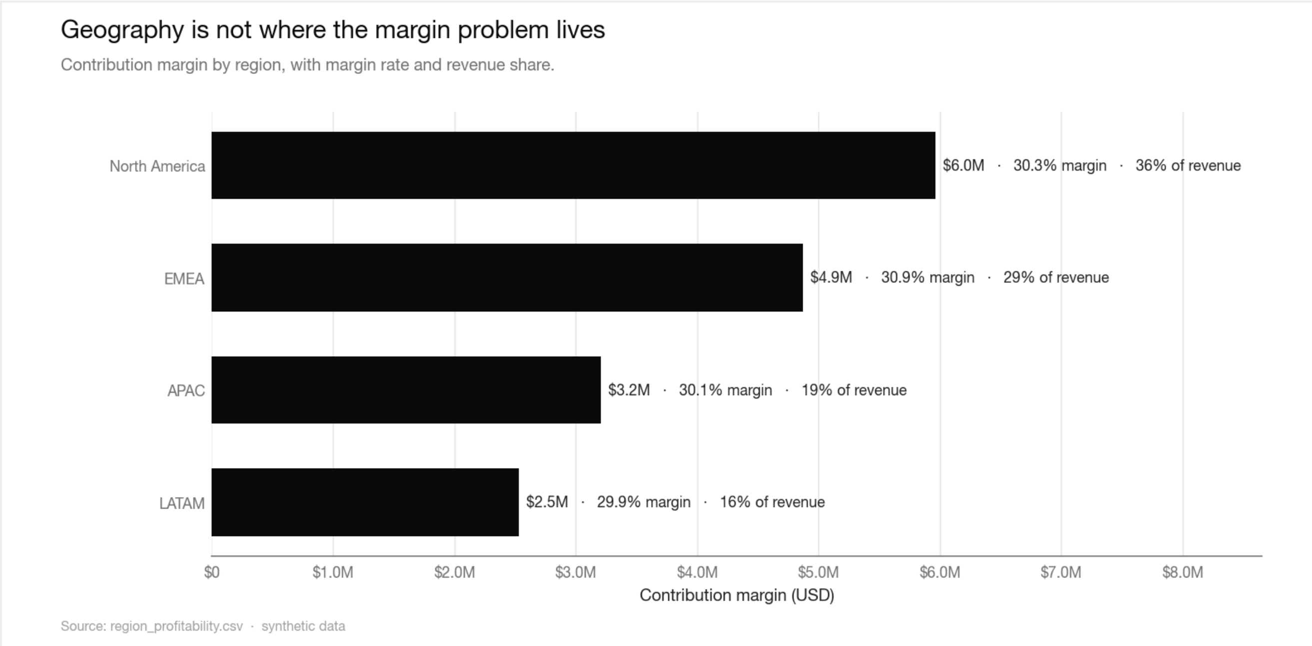

Figure 12. Contribution margin by region with margin rate and revenue share. North America leads on dollars, but every region sits within a point of the others on rate. Geography is a scale story, not a margin story.

The product view sharpens the diagnosis. Services run at just 17.7% margin on $13.6M of revenue (25% of the total), the clear low-margin growth pocket. Premium runs at 24.9%, while Core and Add-on, the two highest-rate lines, run at 41.5% and 47.6%. The spread between the best and worst product line is nearly 30 margin points.

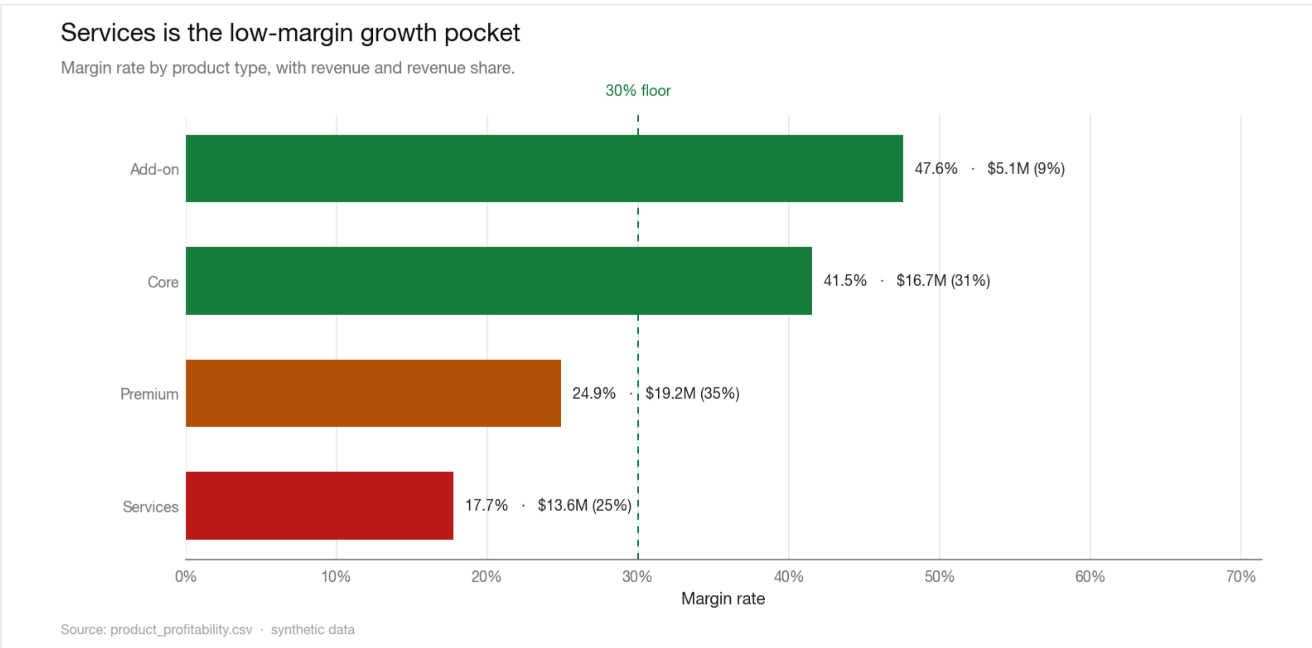

Figure 13. Margin rate by product type with revenue. Services is the only major line well below the floor, and it is large enough to drag the blended rate.

Taken together, the segment and product views say the margin problem is a mix and cost-to-serve problem, not a pricing-everywhere problem. The fix is targeted: re-price or re-scope the Services line and the Enterprise delivery model, rather than raising prices across a base that is already running at or above the floor everywhere else.

5.7 Revenue is concentrated in a thin band of customers

Revenue is far from evenly spread. The top 10% of customers account for 64% of total revenue, the top 20% for 81%, and the top 1% alone for 18%. The Gini coefficient of lifetime revenue is 0.77, well into the territory where a small group of accounts carries the business.

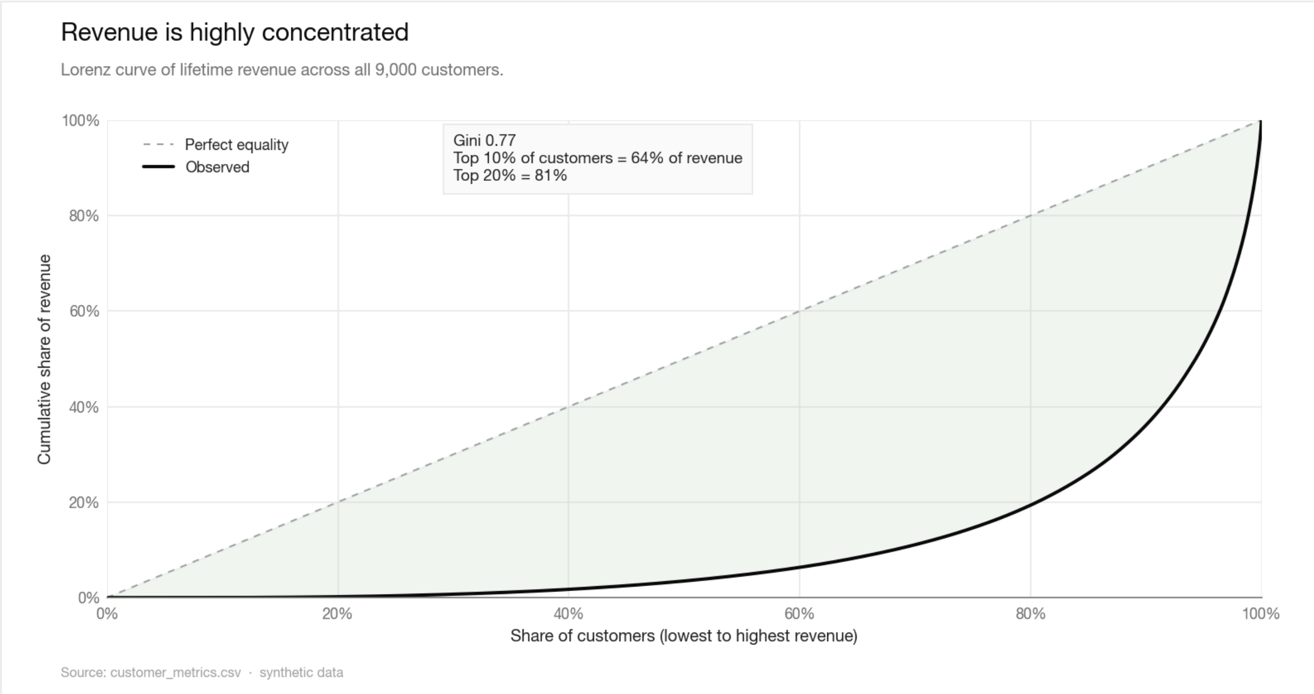

Figure 14. Lorenz curve of lifetime revenue across all 9,000 customers. The deep bow away from the equality line is the concentration: a fifth of customers hold four fifths of revenue.

The distribution behind the curve is a long right tail. Median lifetime revenue is $1,362 while the mean is $6,066, more than three times higher, because a small number of customers reach up to $266K each. The 843 customers who never transacted sit at the bottom of the same distribution and pull the median down further.

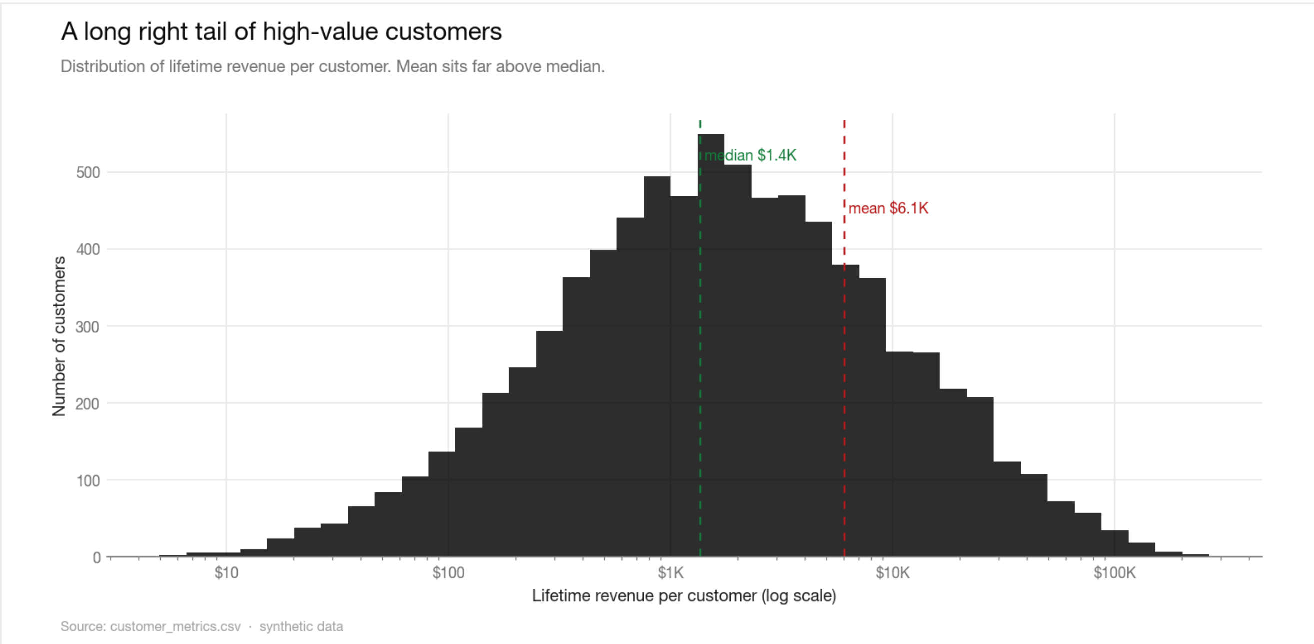

Figure 15. Distribution of lifetime revenue per customer on a log scale. The mean sits far to the right of the median, the signature of a heavy tail.

Concentration interacts directly with the retention finding. Because value pools in a thin band, the cost of early decay depends on which customers decay. Tenure is a real driver of revenue, with a Pearson correlation of 0.42 between lifetime days and lifetime revenue and 0.71 between transaction count and revenue, so customers who survive the first six months and keep transacting are disproportionately the ones who matter.

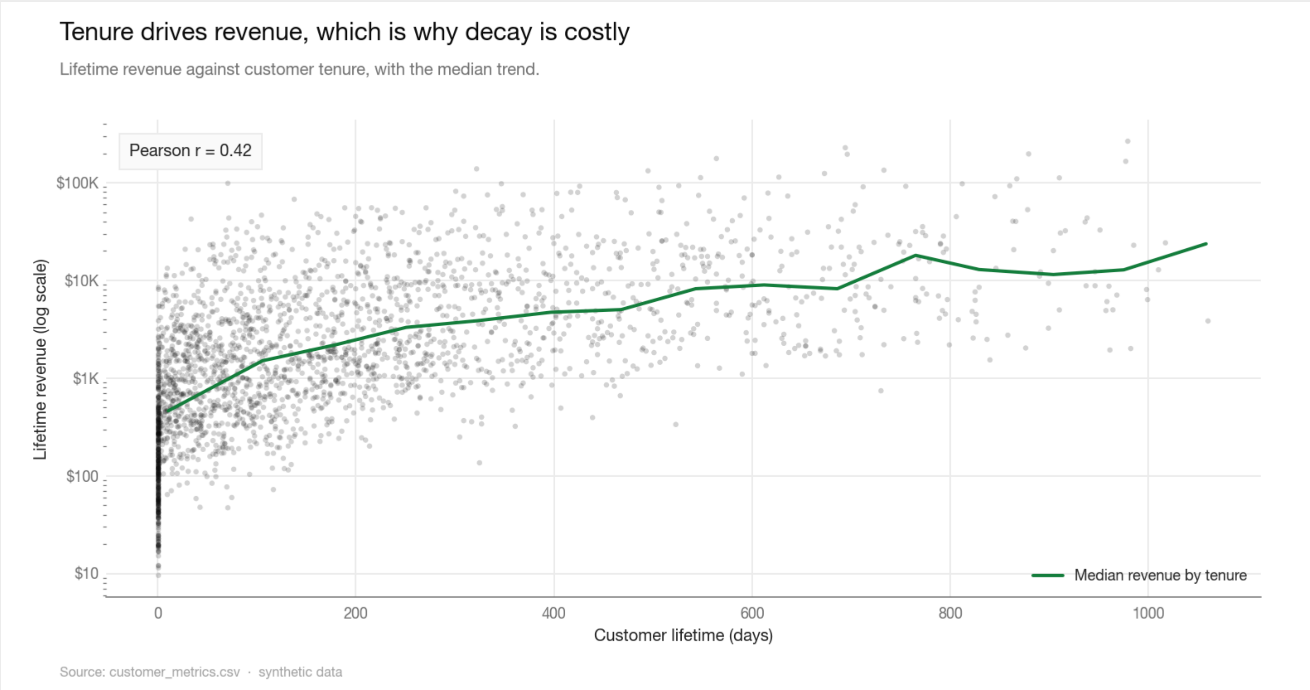

Figure 16. Lifetime revenue against tenure, with the median trend. Longer-tenured customers are worth materially more, which is precisely why losing them early in Figure 6 is expensive.

The implication for the budget is direct. Concentration plus tenure-driven value means the business should weight acquisition toward channels that bring durable, high-value customers, and should protect the high-value base it already has. Both point the same way as the channel economics in Section 5.5.

5.8 Reallocation adds $7.9M with no extra budget

The reallocation scenario scales organic, referral, and partners within guardrails, cuts paid search and social ads by 35%, holds email, and reinvests the freed budget into the efficient channels up to a 100% scale-up cap per channel. Total budget is unchanged, so every dollar of uplift is a pure allocation effect.

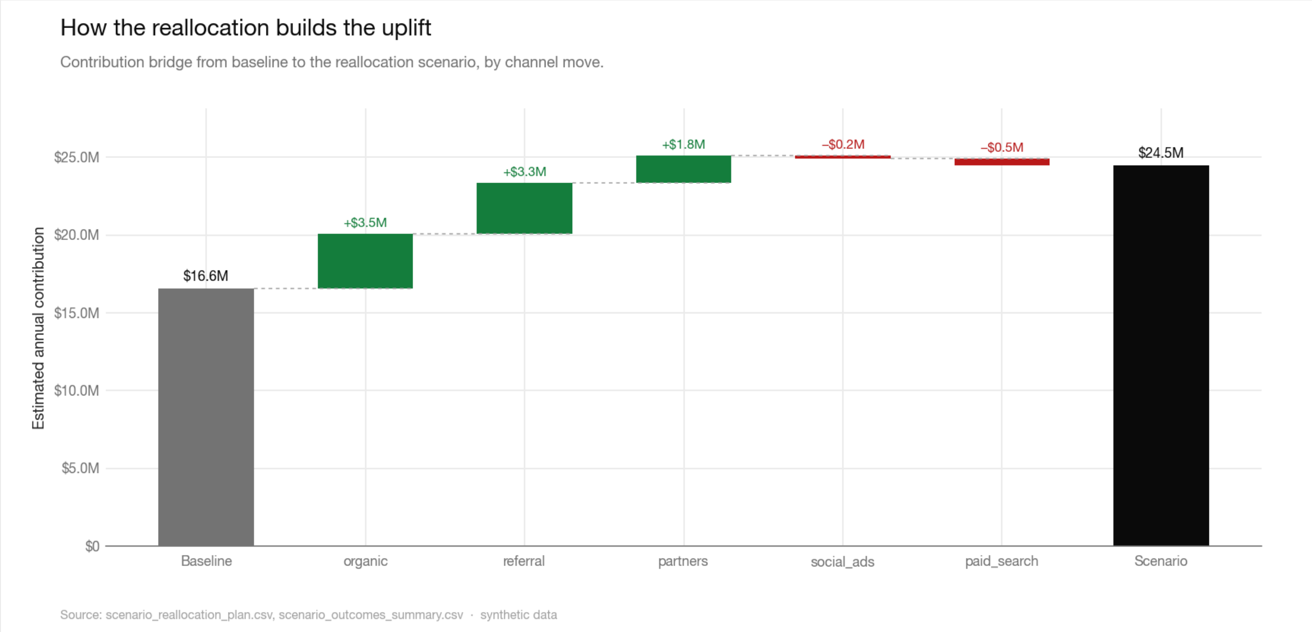

Figure 17. Contribution bridge from baseline to scenario by channel move. Scaling the three efficient channels adds the bulk of the uplift; trimming the two inefficient ones removes a small contribution drag.

CHANNEL	STATUS	BASELINE SPEND	SCENARIO SPEND	SPEND CHANGE	CONTRIBUTION CHANGE
organic	**efficient**	$257,299	$514,599	+100%	+$3,519,556
referral	**efficient**	$478,649	$957,299	+100%	+$3,288,769
partners	**efficient**	$638,558	$1,226,718	+92%	+$1,750,014
email	**borderline**	$394,559	$394,559	+0%	+$0
social_ads	**inefficient**	$1,779,919	$1,156,947	-35%	–$187,022
paid_search	**inefficient**	$2,003,250	$1,302,112	-35%	–$457,948

Estimated annual contribution rises from $16.6M to $24.5M in the base case, an uplift of $7.9M, or 48%. The result is robust to the elasticity assumptions. The scenario is stress-tested across a best, base, and worst case, with CAC and LTV moved against the policy in the worst case.

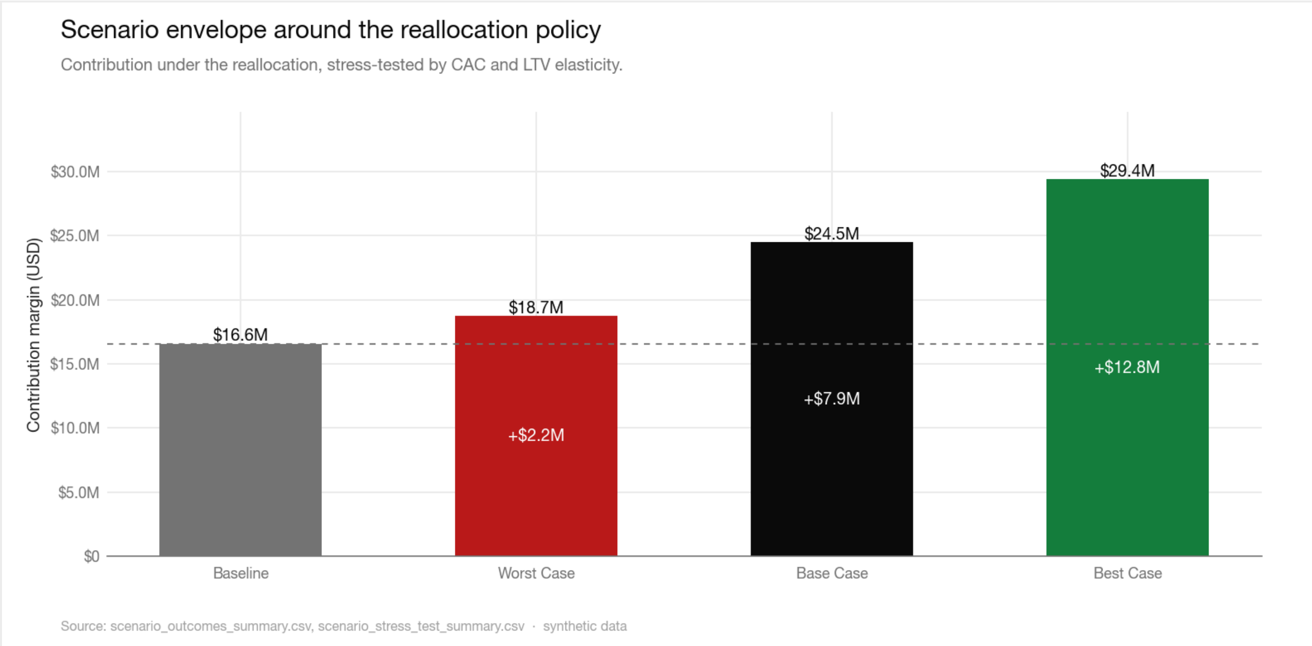

Figure 18. Contribution under the reallocation across stress cases. Every case clears the baseline; even the worst case, with CAC up 15% and LTV down 12% on the scaled channels, adds $2.2M.

SCENARIO	CAC MULT.	LTV MULT.	CONTRIBUTION	UPLIFT VS BASELINE
Best case	0.90x	1.08x	$29,372,878	+$12,808,848
Base case	1.00x	1.00x	$24,477,399	+$7,913,368
Worst case	1.15x	0.88x	$18,730,531	+$2,166,500

The worst case, with CAC inflated 15% and LTV cut 12% on the channels being scaled, still returns $2.2M above baseline. The policy is also stable across random draws. Repeating it across 5 deterministic seeds, the uplift averages $8.0M with a standard deviation of $100K, a coefficient of variation of 1.3%, and it is positive in 5 of 5 seeds. The range runs from $7.8M to $8.1M.

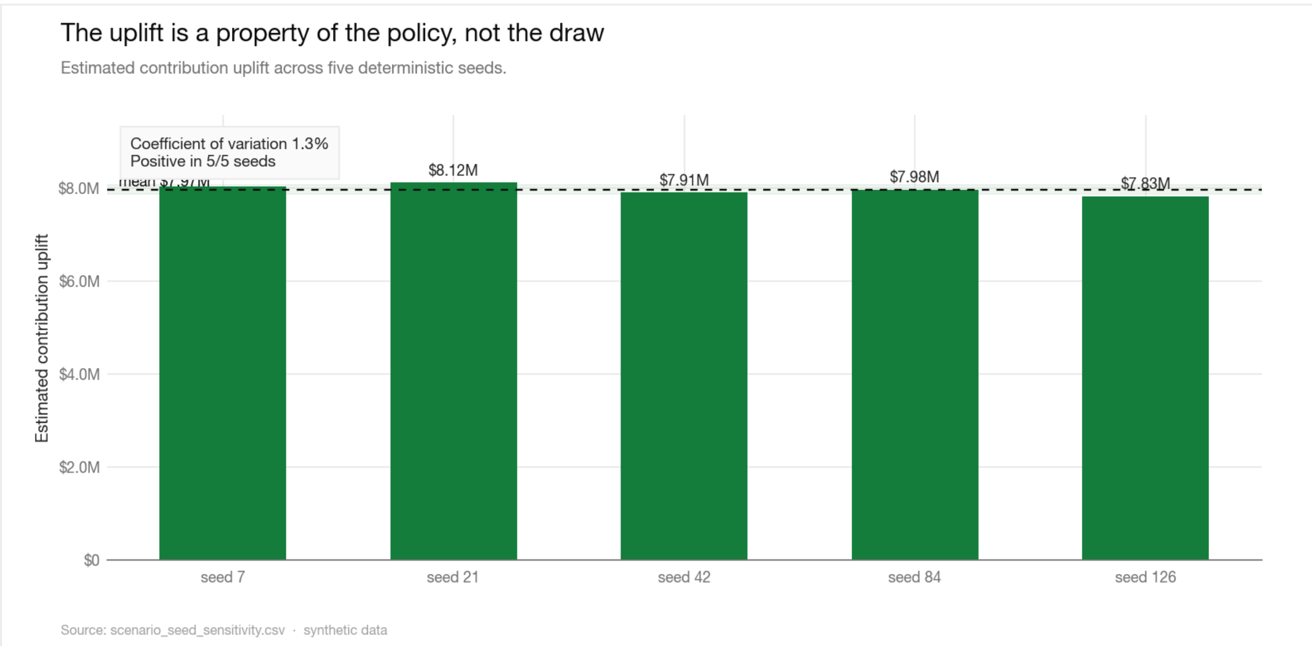

Figure 19. Estimated uplift across five deterministic seeds. The tight band around the mean shows the outcome is a property of the allocation, not of one favourable draw.

SEED	BASELINE CONTRIB.	SCENARIO CONTRIB.	UPLIFT	TOP SCALE-UP	TOP CUT
7	$16,931,189	$24,965,691	+$8,034,503	partners	paid_search
21	$16,575,274	$24,695,403	+$8,120,129	partners	paid_search
42	$16,564,031	$24,477,399	+$7,913,368	partners	paid_search
84	$16,415,282	$24,390,470	+$7,975,188	partners	paid_search
126	$16,571,098	$24,398,607	+$7,827,509	partners	paid_search

6 Risks, limitations, and caveats

The findings are only as good as the assumptions behind them. This section states the limits plainly so the recommendations in Section 7 are read with the right confidence.

The data is synthetic

Every figure in this report comes from a synthetic dataset. It is internally consistent and passes all 11 validation checks, which makes it suitable for demonstrating method and for relative comparison between channels, segments, and cohorts. It is not a forecast of any real market, and the absolute dollar figures should not be read as predictions. The structure of the findings, the direction of the channel split, and the mechanics of the reallocation are the transferable part.

Observed LTV is window-limited

Lifetime value is measured over the available window, not over a true customer lifetime. Channels that acquire younger cohorts have had less time to accumulate LTV, which can understate their true value. The efficiency gap between channels is large enough (20× versus 0.43×) that a window correction would not flip the classification, but the exact ratios would move.

Attribution is single-touch by channel

Each customer is assigned to one acquisition channel, and spend is allocated at the period level. Real customer journeys cross channels, and lagged or assist effects are not modelled. A channel that looks inefficient on last-touch economics could be assisting conversions credited elsewhere. The reallocation is bounded and reversible partly to absorb this risk.

The decomposition depends on the mix dimension

The growth decomposition uses segment as the mix dimension. Defining mix by region, product, or channel would allocate the 6% mix term differently. The volume-led conclusion is robust because volume dominates at 69% regardless of how the smaller mix term is split, but the mix figure itself is dimension-dependent.

Elasticities are assumptions, not measurements

The scenario applies assumed CAC and LTV elasticities rather than measured ones. The stress test and seed sweep are there precisely because the elasticities are uncertain. The honest read is that the uplift is positive across every case tested, not that its central value of $7.9M is precise. Treat the worst case of $2.2M as the planning floor.

Cohort maturity

Long-horizon retention reads draw only on cohorts old enough to have reached that age. The month 24 figure rests on fewer cohorts than the month 6 figure, so the tail of the retention curve is noisier than its head. The month 3 to 6 decay, which carries the recommendation, sits in the best-observed part of the curve.

7 Recommendations and action priorities

Growth is real but expensive. The business is buying volume through channels that lose money, keeping the margin rate pinned at the 30% floor, and leaning on continued spend that fast cohort decay makes hard to sustain. The fix does not require more budget. It requires moving the budget that already exists toward the customers worth acquiring and keeping. The four recommendations below are ordered by value and urgency.

1. PRIORITY 1 **Reallocate spend away from paid search and social ads this quarter.**
   Both return under 0.93× cost and together hold 68% of acquisition spend. Cut each by 35% and redirect the freed $1.3M into organic, referral, and partners, holding each to its LTV/CAC and payback guardrails. This is the 48% contribution uplift modelled in Section 5.8, worth $7.9M in the base case and at least $2.2M in the worst case. It is the only recommendation here that adds contribution without spending a dollar more.

2. PRIORITY 2 **Fix month 3 to 6 retention before slowing acquisition.**
   The steepest decay is early-life, with median revenue retention falling from 88% at month 3 to 68% at month 6. Target onboarding and first-value delivery for the segments and channels feeding the most volume, and track month-6 revenue retention as the success metric. Because revenue is concentrated (81% in the top 20% of customers), prioritise retention work on the high-value tail identified in Section 5.7.

3. PRIORITY 3 **Address the Services and Enterprise margin gap with pricing and cost-to-serve, not volume.**
   Services at 17.7% and Enterprise at 26.2% are where mix dilutes the blended rate. Re-price or re-scope the lowest-margin product and delivery slices rather than growing into them. Leave the segments and regions already at or above the 30% floor alone; the problem is targeted, not systemic.

4. PRIORITY 4 **Hold email flat and run a focused economics experiment.**
   Email is borderline at 2.63× with a 14-month payback. Do not scale or cut it on the current evidence. Run CAC and payback experiments for one or two quarters and decide its fate on the result. Apply the same guardrails to the efficient channels as they scale, since the scenario assumes diminishing returns and the rollout should respect them.

> The reallocation is the priority. It adds $7.9M of contribution without spending a dollar more and stays positive in every case tested. The other three recommendations protect and compound that gain.

Sequencing

Start the reallocation immediately, since it is budget-neutral and reversible. Begin the retention work in parallel, because it determines how much of the reallocated spend converts into durable value. Treat the margin re-pricing as a quarter-two action that needs commercial and finance alignment, and the email experiment as a continuous test that informs the next budget cycle.

Appendix

A. Channel unit economics (full)

CHANNEL	CUSTOMERS	SPEND	CAC	AVG LTV	MED LTV	LTV/CAC	PAYBACK (MO)	STATUS
organic	1,736	$257,299	$148	$3,014	$984	20.33	1.8	efficient
referral	1,668	$478,649	$287	$2,931	$1,117	10.21	3.5	efficient
partners	889	$638,558	$718	$3,127	$1,085	4.35	8.3	efficient
email	455	$394,559	$867	$2,282	$742	2.63	13.7	borderline
paid_search	2,338	$2,003,250	$857	$797	$221	0.93	38.7	inefficient
social_ads	1,914	$1,779,919	$930	$398	$98	0.43	84.2	inefficient

B. Segment and region profitability

SEGMENT	CUSTOMERS	REVENUE	CONTRIBUTION MARGIN	MARGIN RATE	REVENUE SHARE
Enterprise	779	$26,282,659	$6,890,020	26.2%	48.1%
Mid-Market	2,415	$20,220,544	$6,814,179	33.7%	37.0%
SMB	3,184	$6,288,792	$2,232,644	35.5%	11.5%
Startup	2,622	$1,803,972	$627,187	34.8%	3.3%

REGION	REVENUE	CONTRIBUTION MARGIN	MARGIN RATE	REVENUE SHARE
North America	$19,715,177	$5,964,048	30.3%	36.1%
EMEA	$15,778,734	$4,868,825	30.9%	28.9%
APAC	$10,638,073	$3,203,088	30.1%	19.5%
LATAM	$8,463,982	$2,528,069	29.9%	15.5%

C. Product profitability

PRODUCT TYPE	REVENUE	CONTRIBUTION MARGIN	MARGIN RATE	REVENUE SHARE
Add-on	$5,142,972	$2,446,225	47.6%	9.4%
Core	$16,729,012	$6,945,826	41.5%	30.6%
Premium	$19,150,210	$4,764,307	24.9%	35.1%
Services	$13,573,772	$2,407,672	17.7%	24.9%

D. Monthly revenue health (full series)

The 36-month series behind Figures 1, 2, 3, and 4.

MONTH	REVENUE	CONTRIBUTION MARGIN	MARGIN RATE	ACTIVE CUST.	TRANSACTIONS	REV. GROWTH
Jan 2023	$65,387	$23,197	35.5%	67	126	

Feb 2023	$160,455	$51,287	32.0%	125	233	+145.4%
Mar 2023	$233,272	$79,958	34.3%	193	331	+45.4%
Apr 2023	$355,715	$102,498	28.8%	265	447	+52.5%
May 2023	$417,188	$117,628	28.2%	329	543	+17.3%
Jun 2023	$524,416	$147,813	28.2%	396	715	+25.7%
Jul 2023	$522,944	$157,436	30.1%	431	709	-0.3%
Aug 2023	$523,311	$157,384	30.1%	476	819	+0.1%
Sep 2023	$684,967	$206,665	30.2%	526	960	+30.9%
Oct 2023	$783,936	$227,396	29.0%	550	976	+14.4%
Nov 2023	$923,082	$271,152	29.4%	632	1,126	+17.7%
Dec 2023	$1,018,738	$305,513	30.0%	671	1,206	+10.4%
Jan 2024	$869,209	$264,527	30.4%	714	1,251	-14.7%
Feb 2024	$1,172,528	$349,835	29.8%	770	1,400	+34.9%
Mar 2024	$1,125,870	$353,665	31.4%	892	1,570	-4.0%
Apr 2024	$1,162,940	$354,161	30.5%	963	1,659	+3.3%
May 2024	$1,429,033	$425,027	29.7%	1,017	1,792	+22.9%
Jun 2024	$1,409,157	$440,333	31.2%	1,023	1,862	-1.4%
Jul 2024	$1,551,185	$476,109	30.7%	1,069	1,875	+10.1%
Aug 2024	$1,648,498	$505,093	30.6%	1,134	2,014	+6.3%
Sep 2024	$1,770,296	$538,881	30.4%	1,170	2,048	+7.4%
Oct 2024	$1,804,459	$557,671	30.9%	1,244	2,309	+1.9%
Nov 2024	$1,703,617	$539,287	31.7%	1,266	2,268	-5.6%
Dec 2024	$1,911,900	$599,972	31.4%	1,392	2,434	+12.2%
Jan 2025	$2,032,353	$606,116	29.8%	1,404	2,503	+6.3%
Feb 2025	$2,076,299	$626,984	30.2%	1,499	2,583	+2.2%
Mar 2025	$2,416,792	$719,595	29.8%	1,649	2,974	+16.4%
Apr 2025	$2,466,828	$732,700	29.7%	1,725	3,109	+2.1%

May 2025	$2,346,196	$706,402	30.1%	1,779	3,180	-4.9%
Jun 2025	$2,641,011	$781,949	29.6%	1,839	3,226	+12.6%
Jul 2025	$2,658,000	$817,483	30.8%	1,916	3,429	+0.6%
Aug 2025	$2,693,582	$802,697	29.8%	1,943	3,435	+1.3%
Sep 2025	$2,767,966	$841,698	30.4%	1,963	3,488	+2.8%
Oct 2025	$2,868,488	$898,570	31.3%	2,010	3,686	+3.6%
Nov 2025	$2,732,280	$833,462	30.5%	2,000	3,637	-4.7%
Dec 2025	$3,124,066	$943,889	30.2%	2,245	4,027	+14.3%

E. Cohort retention curve (months 0 to 24)

Median retention across cohorts at each age, with the count of cohorts mature enough to be observed. The series behind Figures 6 and 7.

MONTHS SINCE SIGNUP	ACTIVITY RETENTION	REVENUE RETENTION	COHORTS OBSERVED
0	100.0%	100.0%	36
1	97.3%	91.6%	35
2	85.9%	81.4%	34
3	81.7%	88.4%	33
4	74.5%	75.5%	32
5	70.2%	67.6%	31
6	65.1%	68.3%	30
7	58.5%	56.0%	29
8	56.4%	63.8%	28
9	53.3%	55.3%	27
10	48.3%	51.1%	26
11	43.3%	47.6%	25
12	42.4%	49.6%	24
13	42.1%	42.7%	23
14	36.2%	44.5%	22
15	33.3%	37.1%	21
16	32.2%	34.9%	20
17	29.6%	34.5%	19

18	29.5%	33.9%	18
19	26.7%	34.9%	17
20	27.1%	26.2%	16
21	24.0%	29.0%	15
22	21.9%	22.4%	14
23	21.6%	23.1%	13
24	19.7%	21.8%	12

F. Data validation gate

All 11 checks run on the raw tables before any analysis. A failure blocks the pipeline.

CHECK	STATUS	DETAIL
table_row_count_non_zero	PASS	customers=9,000, transactions=69,950, marketing_spend=6,576
schema_columns_match	PASS	customers=['customer_id', 'signup_date', 'segment', 'region', 'acquisition_channel']; tran
null_values_raw_tables	PASS	total_raw_nulls=0
grain_key_uniqueness	PASS	duplicate customer_id=0, duplicate transaction_id=0, duplicate date+channel=0
transaction_customer_referential_integrity	PASS	orphan_transaction_rows=0
transaction_date_not_before_signup	PASS	rows_with_transaction_before_signup=0
value_range_sanity	PASS	non_positive_revenue=0, negative_cost=0, negative_spend=0
negative_margin_transaction_review	PASS	rows_with_cost_above_revenue=258, share=0.37%, review_threshold=1.00%
channel_domain_alignment	PASS	customer_channels=['email', 'organic', 'paid_search', 'partners', 'referral', 'social_ads'
marketing_date_channel_coverage	PASS	expected_pairs=6576, observed_pairs=6,576, missing_pairs=0
date_coverage_observed	PASS	customers=2023-01-01..2025-12-31, transactions=2023-01-03..2025-12-31, marketing=2023-01-0

G. Data dictionary and reproducibility

FIELD	DEFINITION
contribution_margin	Revenue minus direct delivery cost
average_LTV	Mean cumulative contribution margin per acquired customer, including zero-transaction customers
median_LTV	Median cumulative contribution margin per acquired customer
CAC	Period channel spend divided by customers acquired in the channel
LTV_to_CAC	average_LTV divided by CAC; efficient at ≥ 3.0
approximate_payback_period	Months of contribution required to recover CAC

median_revenue_retention	Median cohort revenue at month m relative to month 0
Gini coefficient	Concentration of lifetime revenue across customers; 0 is even, 1 is fully concentrated